



Enhancing Human-Robot Collaborative Search Through Efficient Space Sharing with On-Demand Bidirectional Interaction

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Abstract. Human-robot collaborative search tasks represent a growing field of research, focusing on scenarios where humans and robots coexist in a shared space with the objective of locating specific objects. The complexity of these tasks lies not only in the physical search itself but also in the efficient sharing of the search space between human and robot participants. This paper addresses the critical aspect of communication within human-robot teams, specifically the timing and content of inquiries regarding areas already searched or planned to be searched by each party. We propose innovative methods aimed at optimizing the decision-making process for robots on when to initiate dialogue and how to effectively manage inquiries from human counterparts, who may pose questions at unpredictable times. We develop a system that facilitates such an on-demand interaction for evaluating the proposed communication strategies, and conduct experiments within a virtual environment designed to mimic real-world conditions. In this simulated space, human participants interacted with the robot through avatars, enabling a controlled yet realistic examination of human-robot interaction dynamics. Our experimental results highlight the significant impact of effective dialogue on the efficiency of collaborative search tasks. We demonstrate that timely and relevant exchanges between humans and robots can substantially reduce search times and improve overall task performance. This study not only advances our understanding of human-robot collaboration but also lays the groundwork for future developments in communication protocols for cooperative endeavors involving humans and robotic systems.

1 Introduction

Recently, mobile service robots have become increasingly popular for assisting people in various tasks, including delivery, care, and lifestyle support. A key role for these robots is in lifestyle support, where a frequent task involves locating and retrieving objects specified by users. This task has become a standard challenge in many robotics competitions [19]. Another vital area of focus is how

humans and robots interact. Both can work together, either in person or online, to accomplish a task. This paper explores scenarios where humans and robots coexist in the same space physically and collaborate to find a specific item.

One method of collaboration involves the human directing the robot through commands or advice. Various systems enable humans to aid in the robot’s recognition processes or decision-making [17, 18, 21]. For this to work effectively, the human needs to understand the robot’s capabilities to provide suitable instructions. Alternatively, robots can support humans by referencing a model of the task at hand or the human’s condition to determine how and when to offer assistance [7, 13, 22], making the roles of humans and robots in the search task complementary.

Collaborative object search is also seen as a type of distributed search. When robots work together, they can easily and instantly share data over a network, knowing where others have searched or plan to search to optimize their search patterns (e.g., [6]). Dalmasso et al. [4] proposed a shared task representation for human-robot collaborative navigation, and the agents (humans and robots) exchange information via the representation.

However, such direct data exchange is not possible in human-robot collaborations. Instead, the robot must gather this information through observation and dialogue, similar to how humans collaborate. It is also necessary for the human to gather information from the robot in a similar manner. One important notice is that robots and humans can directly communicate with each other only when they are mutually visible, similar to human-to-human collaborative search.

In human-to-human collaboration, search efforts are often conducted independently, with dialogue reserved for when sharing information is beneficial. This presents challenges in determining the right questions to ask [5, 20] and the optimal time to ask them [2, 26], issues we address in the context of human-robot collaboration. The rise of dialog systems and language-based command interfaces for robots, powered by large language models, marks a significant advancement in this field (e.g., [15, 16]). However, their primary focus is on the content of language-based communication, not on the dynamics of on-demand interactions.

In our previous research [12], we developed a method to enhance human-robot cooperative search by having robots monitor human actions to estimate and predict their search paths. This approach enabled robots to initiate conversations when needed, aiming to share the search area more effectively. Our findings suggested that observing human behaviors could improve search efficiency, but our study only explored one-way communication, from the robot to the human, and did not fully assess the impact of engaging in dialogue. In this study, we build upon our earlier work by incorporating two-way communications, allowing both humans and robots to inquire about each other’s search routes at any time.

The goal of this paper is to examine if bidirectional dialogues can further improve search efficiency. Our goal is to demonstrate that interactive dialogues can further optimize search efficiency.

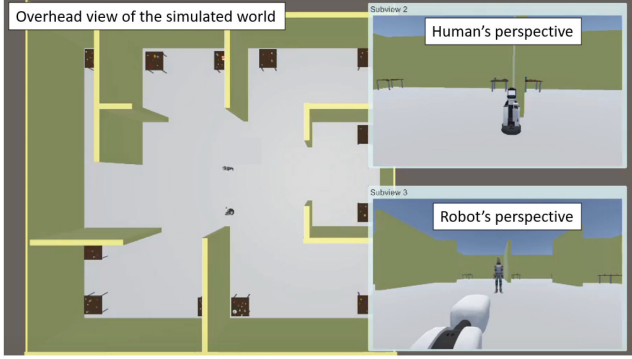


Fig. 1. A snapshot of the simulated environment.

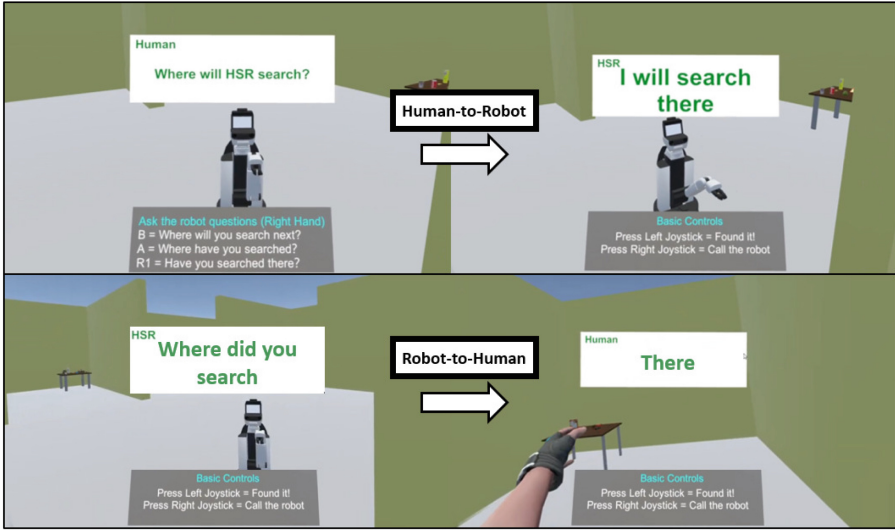


Fig. 2. Interaction with the robot from the human subject's point of view.

This paper offers two main contributions. Firstly, we introduce a robot capable of engaging in two-way conversations with humans, thereby enhancing the coordination of search efforts in tasks that require finding objects. Secondly, we evaluate our method with actual human participants in a simulated environment to determine its effectiveness in promoting human-robot collaboration. We will discuss potential improvements to the interaction method based on the analysis of the experimental results.

Figure 1 shows a simulated environment built using SIGVerse [11], which incorporates a head-mounted display, enabling participants to enter the simulated world as avatars and interact with the robot. The experimental robot

employed is a simulated version Toyota’s Human Support Robot (HSR) [29]. Figure 2 shows example communication scenes in the implemented system. The human-to-robot inquiry is shown at the top, while the robot-to-human inquiry is shown at the bottom.

2 Related Work

2.1 Assistive Robots

Assistive robots help humans in various contexts. One of the essential characteristics of such robots is the timeliness of assistance. Hanai et al. [9] developed a humanoid robot that can assist humans in pick-and-place tasks. By predicting the next actions of humans, the robot can timely generate collision-free hand motions. Goto and Miura [7] developed a robot that can provide timely supporting actions for collaborative assembly based on the assembly instruction. Rasch et al. [23] developed a system of mobile and ambient agents that allow the user to easily issue commands by gesture for tidying up a room. In these works, recognizing human intentions is crucial for effective collaboration. Xie et al. [28] proposed a method for estimating the intentions behind human search actions, using these insights to enable the robot to perform complementary actions. Huang and Mutle [10] proposed anticipatory robot control based on estimating user intentions through gaze patterns. Implicit interaction, shown above, can reduce the human workload and increase the task efficiency. In collaborative object search tasks, where the robot and the human often lose sight of each other, estimating the behavior when they are not visible to each other becomes crucial.

2.2 Human Assistance to Robots

Sprute et al. [27] proposed an object search method with human assistance. A human operator uses multiple smart cameras to monitor both the robot and its environment, suggesting the target object’s location to the robot. Matsushita et al. [18] proposed a similar approach but with an omnidirectional camera interface on the robot. Burks et al. [1] proposed a framework that utilizes human assistance, taking into account the uncertainty in planning optimal robot actions. Although these studies tackle object search problems, they do not include scenarios where humans physically search in the same environment alongside the robot.

2.3 Dialog for Human Assistance

Knepper et al. [14] developed a framework for generating “targeted” requests in natural language, considering common contexts between the robot and the human. Morohashi and Miura [20] developed a method to generate queries that resolve current ambiguities in human commands. Deits et al. [5] developed a system capable of interactively resolving ambiguities in users’ commands by considering probabilistic relationships between symbols and recognized objects.

Shin et al. [25] proposed a framework enabling social robots to generate questions that resolve uncertainties, taking into account knowledge categories, required cognitive processes, and types of questions. In these studies, the robot initiates an inquiry upon observing ambiguity.

Cai and Mostofi [3] proposed a framework for optimizing a human-robot collaborative site surveillance problem, considering the robot's and the human's expected performance and the cost of observation and interaction. Serras et al. [24] developed a dialog system that utilizes the entropy of current information to determine the need for additional information. These studies explicitly address "ask or not to ask" problems and offer solutions for human-robot collaboration involving occasional dialog.

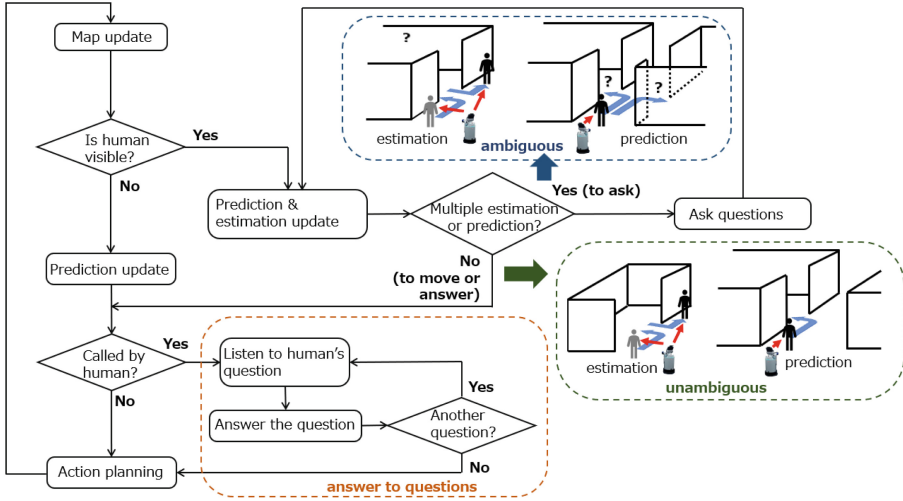


Fig. 3. Diagram for robot's execution. Human past and future search behavior is clear when their potential search areas are visible or obvious, but becomes ambiguous otherwise, prompting the robot to ask the human for clarification. On the other hand, the human can ask questions at any time when they feel it is necessary.

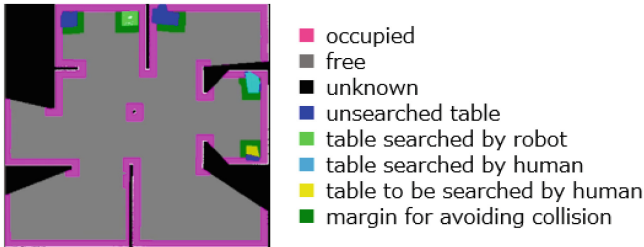


Fig. 4. Labeled map example.

3 Outline of the Proposed Approach

3.1 Problem Setting

In this study, we explore a scenario where a robot and a human collaborate to locate specific objects within an initially unknown environment. The robot employs a SLAM (Simultaneous Localization and Mapping) technique to gradually map the area. Objects of interest are placed on tables; although the total number of these objects is known, their exact positions remain unknown. The robot is equipped to detect and pinpoint the human’s location whenever they come into view. The collaborative search mission is considered complete once all designated objects are found.

3.2 Execution Flow

Figure 3 outlines our proposed method’s process flow. The robot utilizes a labeled 2D grid map to manage environmental data, updating it frame by frame. Upon detecting the human’s presence, the robot assesses areas where the human has likely searched since last sighted and anticipates where they might search next within a set timeframe. To prevent both parties from redundantly searching the same area, the robot inquires about the human’s past or intended search areas if there are several potential regions. If the human is out of view, the robot extends its previous search predictions for a certain period.

Similarly, the human can initiate communication by calling the robot. If called, the robot approaches to engage and address any queries regarding its search strategy.

Following this bidirectional interaction, the robot revises the map and makes its search plan accordingly. Figure 4 provides a visual representation of the map, where each cell’s color shows its status. The robot’s search actions include search on detected tables or conducting frontier-based exploration.

We follow the methods developed in [12] for human path estimation and prediction, as well as robot’s path planning for object search. This approach aims to streamline the search process by facilitating effective communication and coordination between the robot and the human, ensuring that the search for objects is conducted as efficiently as possible.

4 Bidirectional Human-Robot Communication

4.1 Robot-to-Human Inquiries

Initiating an Interaction with the Human. When multiple candidates exist for the human’s estimated or predicted paths, including all in the labeled map and excluding them from the robot’s search destinations can unduly limit the robot’s searchable area, preventing table examinations. Thus, we reduce excessive estimation and prediction candidates by facilitating a dialog between the robot and the human. We follow the idea of ambiguity-driven dialog initiation proposed in [12].

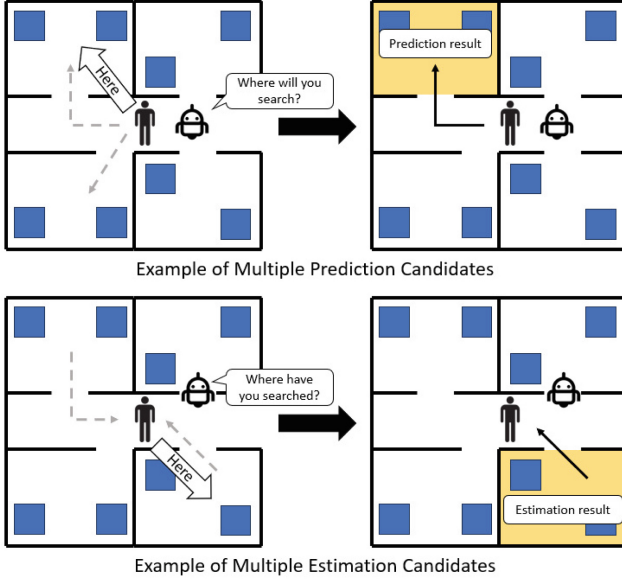


Fig. 5. Examples of ambiguity-driven, robot-generated clarification questions. (top) ambiguous prediction of future search; (bottom) ambiguous estimation of past search.

Generating Questions. We compute the deviation between the elapsed time and travel time for each estimated path. We then compare the closest value to the elapsed time with the second closest, asking “Where did you search?”. if the difference is less than a set threshold. Otherwise, pointing towards the direction of the path closest to the elapsed time, the robot asks “Did you search here?”. For the question “Did you search here?”, we calculate the robot’s pointing direction. The robot’s pointing direction indicates the midpoint of the tables and frontiers’s coordinates on the estimated path.

Similarly, upon obtaining multiple prediction candidates from human behavior prediction, we formulate questions to precisely identify the prediction outcome. Contrary to generating questions for estimation candidates, for prediction candidates, the robot inquires “Where will you search next?” regardless of the candidate paths. An example of this implementation is shown in Fig. 5.

The robot’s question messages are played as audio by the simulation’s robot. Additionally, these messages are displayed as text in the human’s point-of-view footage within the simulation. During the simulation, the robot’s pointing direction is conveyed as it moves its arm towards the intended direction, mimicking a pointing gesture.

4.2 Human-to-Robot Inquiries

In our previous work [12], while robot-to-human dialog effectively reduced duplicate searches, we observed that human subjects still performed some duplicated

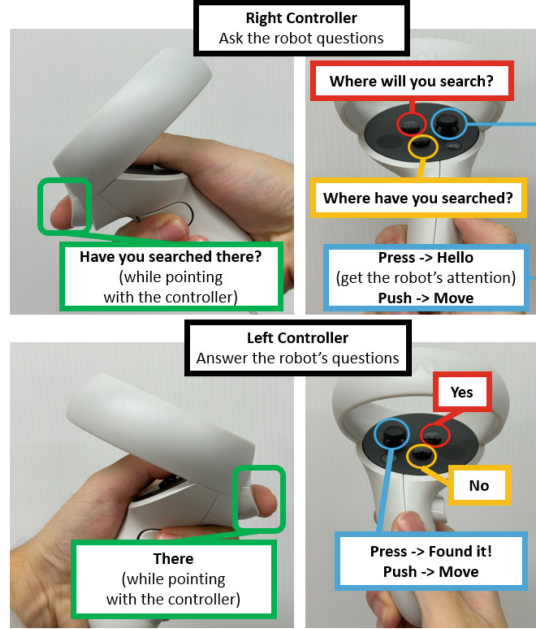


Fig. 6. Controller mapping.

searches, lacking a method to verify previously searched areas without the robot being in their view. Therefore, we introduce human-to-robot dialog, allowing humans to inquire about the robot’s past searches and planned searches, as depicted in the execution flow (see Fig. 3).

To initiate an interaction, the human should first approach the robot to a reasonable distance before signaling for its attention. This distance is defined as 7.6 m, aligning with the outer boundary of public space in human interpersonal distances [8]. Upon entering this distance, the robot will respond and advance towards the human upon receiving a signal.

Likewise, the human can pose three questions to the robot: “Where have you searched?”, “Where will you search?” and, while pointing, “Have you searched there?”. The robot will answer with a pointing gesture indicating its past and future search areas, respectively, and a “Yes” or “No” to the question “Have you searched there?”. The timing and content of the inquiries are entirely left to the discretion of the human.

4.3 Processing and Receiving Input

Human Input. Throughout the collaborative search task, the human participant engages in the task and communicates with the robot via inputs made through the Meta Quest 2 VR headset’s controller. The inputs are categorized into two main parts: the left controller is primarily utilized for responding to the

robot’s queries, while the right controller is mainly used to pose questions to the robot, as shown in Fig. 6. The human uses the controller to point the direction of interest, facilitating more precise communication with the robot.

Processing the Human’s Response to Robot’s Questions. The robot processes the human’s responses to the robot’s questions as follows. In the case of the question, “Did you search here?”, the human can respond in two ways, each of which is processed as follows:

- “Yes”: the candidate on which the question was generated is used as the result of the human path estimation.
- “No”: the candidate on which the question was generated is excluded.

For the inquiry “Where did you search?” the outcome is determined by the direction indicated by the human. This result matches the candidate whose direction best aligns with both the indicated direction and the path from the human to the candidate’s coordinates. Similarly, for the query “Where will you search next?” the selected option is the one that best aligns with the direction indicated by the human and the path from the human to the center of the candidate’s coordinates. An example of this implementation from the human subject’s perspective is shown in the bottom part of Fig. 2.

Processing the Robot’s Response to Human’s Questions. When the robot receives a call of attention from the human, it approaches the human to face them and prepares to receive the question. When the human asks “Where have you searched?” the robot reviews its search history and points towards its most recent search destination in response. When asked “Where will you search?” the robot consults the labeled map (see Fig. 4) and, based on its action plan, points towards its next search location.

When the human points and asks “Have you searched there?” the robot matches the pointing direction with previously searched locations. This process is explained in detail in the following section.

An example of the robot’s response to the human’s question is shown in the upper part of Fig. 2.

4.4 Handling Human Pointing Gestures

Pointing Gesture Matching. When the robot visits any table or frontier, it records and saves the approximate coordinates of the target viewpoint and categorize them as either a *table viewpoint* or a *frontier viewpoint*. In response to the “Have you searched there?” question, we match the angle of the human’s pointing gesture with that of the already traversed viewpoints and areas relative to the human’s current position.

In cases where the robot has one or more table viewpoints saved, it prioritizes table viewpoints over frontier viewpoints due to the latter’s reduced reliability in representing searched areas. If the human subject’s pointing gesture matches

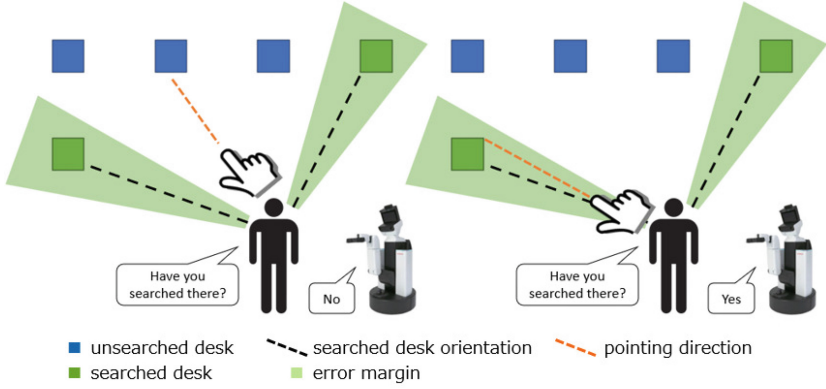


Fig. 7. Example of pointing matching with searched tables.

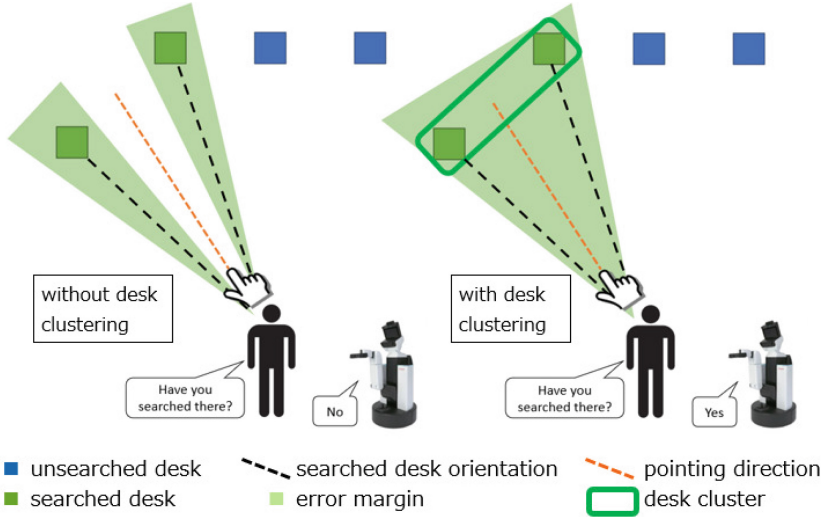


Fig. 8. Example of table clustering.

the angle of the searched table viewpoint, then the robot responds positively to the question with “Yes, I have searched there”. Otherwise, the robot responds negatively with “No, I have not searched there”. A margin of error in the human’s pointing gesture, defined as 20° to both the left and right of the table viewpoint, is also considered. An example of this implementation is shown in Fig. 7.

If there are no table viewpoints, the robot instead matches its previous frontier viewpoints. Similar to the previous situation, if the human subject’s pointing gesture matches the angle of the traversed frontier viewpoint, the robot responds with “Yes, I have searched there”; otherwise, the robot responds with “No, I have not searched there”.

Table Clustering Human pointing gesture can be ambiguous. In this paper, we consider the possibility that the human subject may point to refer to a specific, individual table, or point towards an area to ask whether multiple tables in an area have been searched. To enable the robot to respond to ambiguous cases, we cluster multiple tables together and declare the area between the tables as searched. The maximum size of the table cluster is three, and we define the searched area as the area that is within the outer vertices of each table. An illustration of our implementation of the table cluster is shown in Fig. 8.

When the human subject points towards a table cluster, the robot responds positively with “Yes, I have searched there” if the pointing gesture’s angle falls within the range defined by the cluster’s edges and the human’s current position. With this, the robot responds correctly to searched areas even if the human subject does not point directly at an individual table.

5 Experiments

5.1 Experimental Setup

Participants are instructed to search for three apples in the simulated environment. They are encouraged to respond to the robot’s queries and, at their discretion, may pose questions to the robot when the experimental conditions permit. Participants do not know the shape of the environment beforehand; however, the testing area is neither large nor complex, so they should have no difficulty understanding it shortly.

We compared four methods, each built upon the human movement estimation and prediction method as a basis. Each method underwent testing with every participant, with the following descriptions:

- *No Communication*: No direct communication occurred between the robot and the human subject.
- *Robot-Only Questioning*: Only the robot can pose questions.
- *Human-Only Questioning*: Only the human can pose questions. The robot does not pose questions, even when multiple candidates for estimation and prediction are generated.
- *Bidirectional Communication*: Both the human subject and the robot can pose questions.

The order of presenting the modes was randomized for each participant. Additionally, each participant is given a practice time to become familiar with handling the equipment. One session ends when the robot-human team finds three apples. Both the robot and the human must be close enough to each apple for it to be considered found.

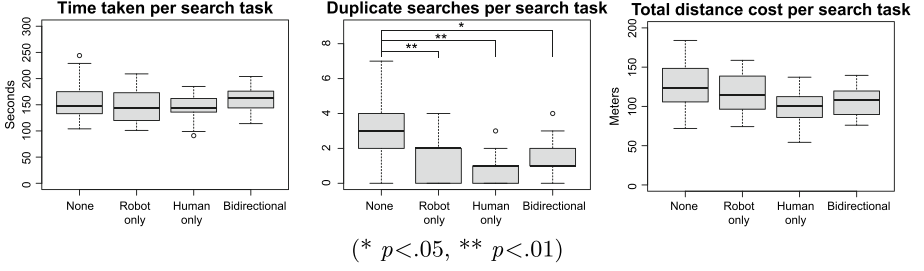


Fig. 9. Experimental results in three criteria.

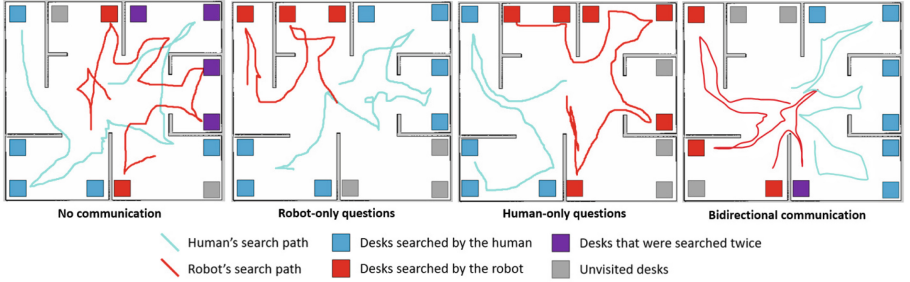


Fig. 10. Example search paths in all conditions.

5.2 Evaluation Metrics

We used the following three evaluation metrics:

- *Total Time Taken per Search Task*: This metric assesses the duration the team requires to locate all target objects.
- *Number of Duplicate Searches*: This metric measures the amount of duplicated searches on tables.
- *Total Distance Traveled by the Human and Robot*: This metric is the cumulative distance covered by both the human subject and the robot throughout the search task.

5.3 Experimental Results

The experiment involved 21 undergraduate and graduate students assessing search efficiency in human-robot interactions across different communication modes. The experiment evaluated the effect of no communication, robot-only, human-only, and bidirectional communication on search times, duplicate searches, and distance costs. Table 1 summarizes the average metric values for the four methods. Introduction of communications enhanced all metrics in robot-only and human-only scenarios, consistent with the expectation of enhanced coordination. These improvements are attributed to strategically using queries to

Table 1. Summary of average metric values.

Communication Type	Search Time [s]	Duplicated searches	Distance travelled [m]
None	156.33	2.95	126.33
Robot-only	148.00	1.43	114.14
Human-only	146.81	0.86	101.13
Bidirectional	159.14	1.48	106.96

avoid revisiting areas, showcasing the effectiveness of communication in enhancing search efficiency. Although bidirectional communication reduced duplicate searches and distance costs, it required slightly more search time compared to the no communication case.

Figure 9 shows the results of the Friedman test followed by Post-hoc Nemenyi’s tests (Holm). For the time taken per search task, the analysis revealed no significant overall effect ($\chi^2(3) = 3.809, p = 0.283, W = 0.181$). There were no significant differences among conditions. For the duplicate searches per search task, the analysis revealed a significant main effect ($\chi^2(3) = 22.733, p < .001, W = 1.083$). Post-hoc test indicated that the mean score for the no communication was significantly higher than others (robot-only: $p < .00$, human-only: $p < .00$, bidirectional: $p < .05$). There were no significant differences for other combinations. For the total distance cost per search task, the analysis revealed a tendency towards significant main effect ($\chi^2(3) = 7.571, p < .10, W = 0.360$). However, the post-hoc test indicated that there were no significant differences.

While the reduction of duplicate searches is significant, that of time taken is not. One reason is that an ambiguity-driven criterion for initiating inquiries may lead the robot to generate more inquiries than necessary. This is also indicated by the results that the human-only strategy got the best scores in all metrics, as humans set inquiries only when needed. This indicates a need to optimize communication strategies for the robot, balancing interaction costs with search efficiency improvements. Another factor is that the present interaction method, using a controller, is less intuitive than human voice-based interactions, thus requiring more time.

The reduction of total distance cost is less significant. This also stems from the present interaction method that necessitates proximity between the robot and the human, increasing both the time and distance traveled. Figure 10 illustrates observed path examples for each of the methods. Introducing communications undeniably reduces duplicate searches, though it sometimes requires the robot and the human to come closer for interaction.

In summary, while there is room to improve the communication efficiency, the introduction of communication holds significant promise for enhancing human-robot collaborative object search.

6 Conclusions and Discussion

This paper presented collaborative object search as a typical problem for human-robot interaction research. We proposed a strategy for efficient sharing of search space between humans and robots, based on mutual observation and anticipation of past and future search behaviors, and on-demand communication for explicit information exchange. We implemented an experimental system using the virtual environment SIGVerse and conducted evaluation experiments with human subjects. The results show that introducing on-demand communications has significant effects in improving collaborative object search performance.

The experimental results have not only shown the potential of on-demand communications in collaborative search but also underscored the need for further improvements and research. To further increase communication efficiency, integrating a voice-based interface could offer a more natural interaction method for humans, potentially making the collaboration process more efficient. Additionally, transitioning to real-world experiments with robots capable of matching human operational speeds could provide deeper insights into practical human-robot collaboration dynamics. Moreover, comparison with human-to-human collaboration scenarios could reveal valuable design considerations for human-robot collaboration scenarios.

Acknowledgment. This work is in part supported by JSPS KAKENHI Grant Number 22H04864.

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